Problem 11.5

An investment account earns 4% during the first year, 5% during the second year, and $X\%$ during the third year. Find X if the three-year average annual interest rate is 10%.

Solution

Let P be the initial amount invested. After three years, using the annual rates earned by the account, its accumulated value is

$$P(1.04)(1.05)(1+X).$$

An average annual interest rate of 10% means that the same initial amount accumulates over three years to

$$P(1.10)^3.$$

Equating the two accumulated values gives

$$P(1.04)(1.05)(1+X) = P(1.10)^3.$$

After canceling P and solving for X, we obtain

$$\begin{aligned} 1+X &= \frac{(1.10)^3}{(1.04)(1.05)}, \\ X &= \frac{(1.10)^3}{(1.04)(1.05)} - 1, \\ X &\approx 0.21886447. \end{aligned}$$

Therefore, the interest rate earned during the third year is

$$\boxed{X \approx 21.8864\%}.$$